

Blades of Chaos

110 mm two-sided printable keychain

Print one A front and one mirrored B back for each finished blade. Both halves sit on their bonding faces with the sculpt facing up. For a pair of blades, print two A halves and two B halves.



Part	File	Per blade
Front half A	blade_A_front.stl	1
Mirrored back B	blade_B_back_mirrored.stl	1
Double-taper alignment key	alignment_key.stl	3

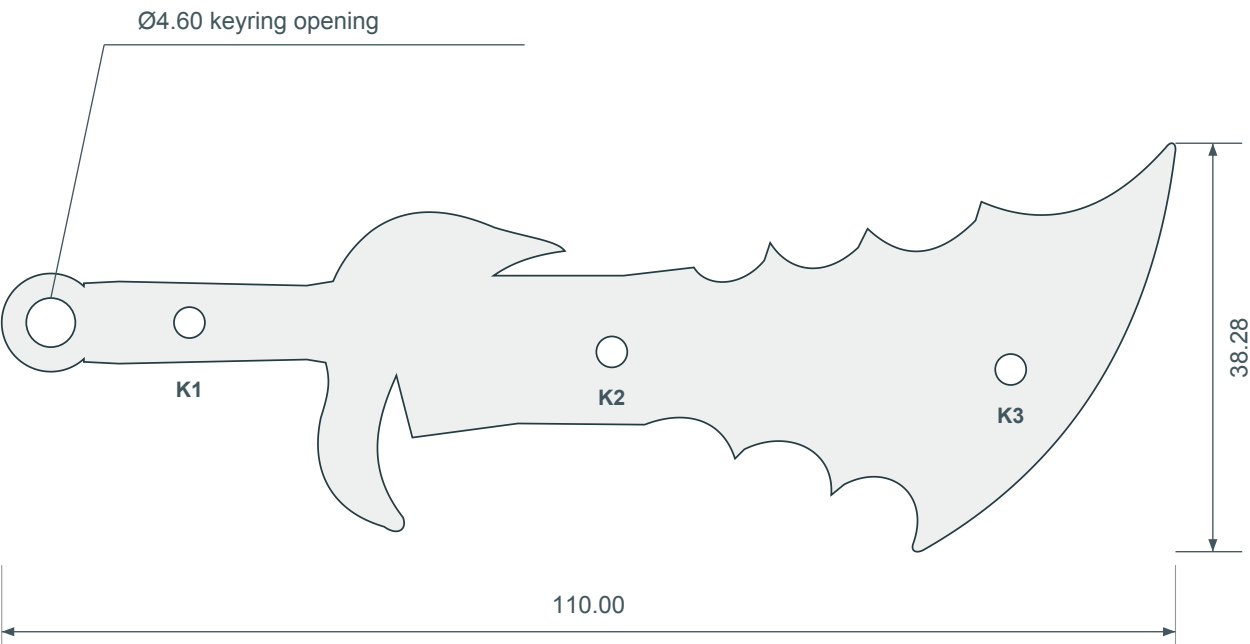
Ready-arranged plates

output_model.3mf / .stl: one A + one B + five keys (two spare).
two_blades_print_plate.3mf: four halves + eight keys (two spare).

Assembled size: 110 × 38.28 × 11.93 mm, including the illustrated 0.08 mm bond line. Keyring hole: Ø4.60 mm. The reference assembly file is for inspection; the arranged plates contain the printing orientations.

Geometry and fit

A HALF - MATING FACE / SOCKET LOCATIONS



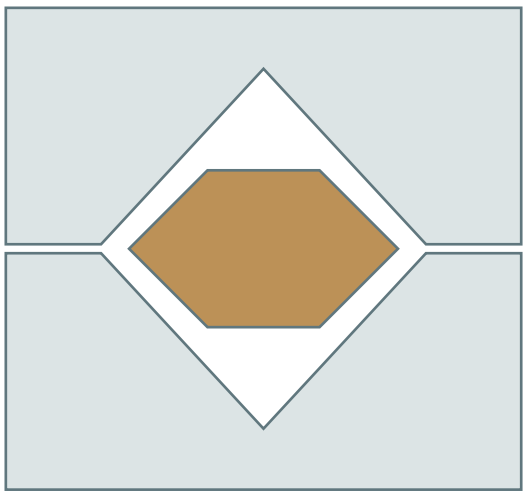
Views are not to scale. Use stated dimensions and supplied 3D geometry.

ASSEMBLED SIDE ENVELOPE



Both external faces carry the sculpt. Center line is the symmetry / joining plane.

SECTION THROUGH ALIGNMENT KEY (ENLARGED)



- 3 identical keys per finished blade
- Key: Ø2.40 max × 1.40 long
- Tip flats: Ø1.00
- Socket mouth: Ø2.90
- Socket depth: 1.566
- Socket roof: 47.2°
- Nominal radial clearance: 0.25
- Preview adhesive gap: 0.08
- Remaining skin: at least 1.06

Bonding faces touch for a flush joint. The 0.08 mm gap is a visualization allowance, not a built-in spacer.

Print, fit and bond



1. Print. Use the supplied orientation, flat bonding face on the bed, supports off. Start with a calibrated 0.25 mm nozzle / 0.10 mm layer profile for the small detail. A 0.4 mm nozzle retains the main shape but softens the fine ornament. Use 4 walls, 5 bottom layers, 6 top layers and about 30% infill.
2. Prepare. Remove any brim from the tiny keys and clean the bonding faces. Keep the conical sockets clear. Lightly flatten a warped seam face before assembly; preserve the socket mouths.
3. Dry-fit. Place three keys in A. Flip B over around its Y axis so its flat face mates with A. Match the pommel hole and outer profile. The keys are loose alignment aids; apply light clamping rather than forcing the fit.
4. Bond. Apply a thin, even layer of adhesive compatible with the printed material to the flat lands, seat B, and clamp until the adhesive manufacturer's full cure time. Clear squeeze-out from the eyelet and outline.
5. Finish. Dress the seam, prime and paint if desired, then fit a metal split ring. For plastic welding, use matching polymers and trial the seam on a coupon; BONDLINE_GAP=0 represents a flush welded joint.

The files pass manifold, winding, mirrored fit, socket skin and print-overhang checks. No physical print, bond-strength test or welding trial has been performed.